Noise Pollution and Human Capital

Armando Meier Felix Schulz

Noise pollution is a widespread environmental externality, but its consequences for human capital formation remain far less studied than those of air pollution or heat. We seek to contribute to the small existing literature on noise and human capital by exploiting the staggered, plausibly exogenous rollout of a nationwide noise-mitigation program in Sweden.

Noise likely deteriorates sleep, well-being, and health

A set of quasi-experimental studies point to causal effects of noise on health. Boes et al. (2013) exploit two unexpected flight-regulation changes in Zurich and find that noise increases sleeping problems and headaches, with naive cross-sectional estimates understating the effect because of sorting. Hener (2022) use daily variation in Frankfurt landing approaches to instrument noise and finds that a 4.1dB increase in background aircraft noise raises the violent crime rate by 6.6 percent, mostly physical assaults on men. Beghelli et al. (2023) exploit an unannounced reallocation of early-morning landings at Heathrow and find that prescription spending on CNS and respiratory medication falls when noise drops; Argys et al. (2020) exploit the FAA's NextGen flight-path concentration and find a 1.6 percentage point increase in low birth weight among mothers close to and downwind of flight paths; and Wang et al. (2022) use a difference-in-differences design around a LaGuardia flight-path change and document increases in insomnia, cardiovascular disease, substance use, and mental-health-related healthcare use — though Li et al. (2021) find no effect from one Dutch policy change, in part because it failed to durably cut noise. Using a novel seismometer-based noise measure, Greenhill (2026) finds that nighttime rail-noise increases from wind-direction shifts worsen infant health, and Lindgren (2021) finds that a Swedish road-noise mitigation program reduced cardiovascular hospitalizations via hypertension. Property values, finally, move in either direction depending on whether noise rises or falls (Allroggen et al. 2025; Moretti and Wheeler 2025) — evidence we return to below, since noise-barrier construction is itself one of our candidate identification strategies.

A larger epidemiological and public-health literature points the same direction but is generally less well identified, relying more on self-reported exposure and on cross-sectional or purely observational variation: noise is linked to worse subjective well-being and life satisfaction (Van Praag and Baarsma 2005; Krekel and Zerrahn 2017), and sleep disturbance from noise is well replicated across studies (Basner and McGuire 2018; Basner 2008; Griefahn et al. 2006; Bodin et al. 2015; Trimmel et al. 2012). Wind-turbine evidence is more mixed: some largely correlational studies find reduced life satisfaction and higher suicide risk (Krekel and Zerrahn 2017; Zou 2017), while Krekel et al. (2023), using a quasi-experimental design, find no health effects at all.

Limited causal evidence on noise and human capital

A small literature studies noise's impact on human capital directly. In this literature, only some studies are able to convincingly account for endogenous self-selection of individuals into neighborhoods and schools in response to noise and rents. To our best knowledge, no study on human capital outcomes has yet successfully separated the effects of noise from co-occurring pollution and traffic exposure. Finally, while

the studies discussed below are exceptionally nuanced in their measurement of cognition, all existing studies measure outcomes only in childhood and no study has attempted to estimate long-run outcomes.

Several studies attempt to identify causal effects in extensive cross-sectional tests of schoolchildren. One study finds worse reading comprehension in a sample selected from schools near Heathrow, UK (Haines et al. 2001). In the larger, multi-country RANCH study, hand-select schools around three major European airports, measure noise and conduct cognitive tests (Clark et al. 2010). They link aircraft noise linearly to worse recognition memory, but also find a positive relationship between road noise and recall. Lercher et al. (2003) run a similar design and find negative effects on memory. Klatte et al. (2017) study primary schoolers around Frankfurt Airport (NORAH) and find that even moderate aircraft noise lowers reading performance. We believe that while sound in measurement, these designs rely too heavily on their control variables to fully identify causal effects.

Dean (2024) shows experimentally that a 7dB increase in workplace noise reduces textile-worker output by about 3 percent and impairs cognitive function, unbeknown to workers. The most convincing existing design is that of Hygge et al. (2002), who exploit the Munich airport relocation as a natural experiment. Testing schoolchildren matched on socioeconomic characteristics before and after the airport move, they find that children newly exposed to aircraft noise show impaired reading and long-term memory, while those who lost exposure show improvements. Klatte et al. (2017) do however note that matching was performed before a significant number of participants were excluded because they did not finish the most difficult test items. Balance after this step was never reported.

We seek to address these gaps by exploiting the staggered, plausibly exogenous timing of Sweden's BoViT insulation and noise-barrier rollout described below.

Literature overview

Table 1: Selected studies on noise pollution, well-being, health, and human capital (full literature discussed in text). Table extends Nyberg (2025). Periods in parentheses are approximate where not directly stated in the source and should be double-checked before circulating further.

Study	Research Question	Methodology & Sample	Main Result	Country (Period)
Moretti and Wheeler (2025)	Are noise externalities capitalized into property prices, and how large is the aggregate cost?	DiD/event study around FDOT noise-barrier construction; hedonic house-price model. FL barrier inventory, 1,143 barriers; CoreLogic transactions, 596,419 home sales in the estimation sample	Noise barriers raise nearby property values by 6.8% within 100m; implies $110bn aggregate U.S. cost, $974/vehicle efficient tax	USA (Florida; barriers 1988–2022, transactions 1990–2022)
Lindgren (2021)	Does a dwelling noise-mitigation program reduce cardiovascular disease?	DiD around a nationwide program; hospitalization/diagnosis registers, 10,113 indiv. who stayed at the same address throughout	-10% cardiovascular disease risk after 7 years (-18–19 cases/1,000), two thirds via hypertension	Sweden (residents 1998–2008; mitigation 1999–2017; outcomes 2003–2017)
Greenhill (2026)	Does nighttime noise harm infant health?	Quasi-exp. using idiosyncratic wind-driven rail-noise variation (share of pregnancy spent downwind); seismometer-based noise + confidential birth-certificate microdata	+2dB nighttime noise → -4% SD infant health index	USA (California; births 1999–2021)
Haines et al. (2001)	Does chronic aircraft noise impair children's reading and mental health?	Quasi-exp. (4 matched high-/4 low-noise schools); group-administered cognitive/mental-health tests + salivary cortisol subsample; 340 children, 8–11y (169 high-/171 low-noise)	Worse reading comprehension; no mental-health or cortisol effect	UK (Heathrow; 1991)
Hygge et al. (2002)	Does a change in aircraft-noise exposure affect children's cognition?	Prospective natural experiment around the Munich airport relocation; 3 testing waves (6mo before, 1yr & 2yr after); 326 children, 8–12y, matched noise/no-noise groups at old & new sites	Reading and long-term memory impaired at newly-exposed site, improved at newly-quiet site	Germany (Munich; 1991–1994)
Clark et al. (2010)	Does aircraft/road noise affect children's cognition, and is the relationship exposure–response (RANCH)?	Multilevel cross-sectional; cued recall, recognition & prospective-memory tests; 2,844 children, 8;10–12;10y, matched schools around 3 airports	Aircraft noise → worse recognition memory (linear); road noise → better cued recall (linear, unexpected direction)	UK, Netherlands, Spain (∼1999–2000)
Klatte et al. (2017)	Does aircraft noise lower reading performance at moderate exposure levels (NORAH)?	Multilevel cross-sectional, continuous exposure–response design (not just hi/lo groups); reading & quality-of-life tests + teacher reports (N=84); 1,243 2nd graders, 29 schools	Even moderate noise (<60dB) lowers reading performance, raises annoyance; no effect on verbal precursors	Germany (Frankfurt; noise May 2011–Apr 2012, testing spring)

continued on next page

Study	Research Question	Methodology & Sample	Main Result	Country (Period)
Dean (2024)	Does workplace noise reduce productivity and cognition, and do workers realize it?	Two randomized field experiments, textile-training course: randomized engine-noise exposure during a sewing task (N=128), plus a willingness-to-pay-for-quiet elicitation	+7dB noise → -3% output, impaired cognition; no WTP premium for quiet (workers unaware of the cost)	Kenya (Nairobi)

Data sources

Table 2: Data sources for exposure, outcome, and auxiliary variables. Cells marked "TBD" need to be confirmed with the data provider before the meeting.

Role in analysis	Dataset (Source)	Level of Detail	Years Covered	Open Access?	Access
Exposure	BoViT project data, incl. dwelling-level noise insulation (Trafikverket)[1]	Dwelling-level, geocoded	TBD – confirm with Trafikverket	No	Request from Trafikverket
Exposure	Sound barriers – roads (Trafikverket)	Barrier-segment level, geocoded (road network); 2,172 segments in the current extract	1950–2025 (51% unrecorded)	Yes	Lastkajen entry 630
Exposure	Sound barriers – railway (Trafikverket)	Barrier-segment level, geocoded (rail network); 1,829 segments in the current extract	1989–2025 (27% unrecorded)	Yes	Lastkajen entry 991
Outcome	National test & final-grade results, school-level aggregates (Skolverket)[2]	School-unit level (not individual, not geocoded itself); grade-point average & national-test pass-rate measures	1998–2019 + 2022/23–2025/26 (gap ∼2020–2022)	Yes	1998–2019: archived SIRIS exports on Skolverket's S3 (via jplus-plus/skolstatistik); 2022/23–: Skolverkets statistikdatabas (PxWeb API)
Outcome	National test & final-grade results, individual-level microdata (Skolverket)	Individual student-level microdata	TBD	No	Microdata request via Skolverket
Outcome	Labor-market register: earnings, employment status (Statistics Sweden, SCB)	Individual-level microdata	1990–present (LISA, confirm exact span)	No	Microdata service (MONA)
Auxiliary	School location register (Skolenhetsregistret, Skolverket)	School-level, geocoded (address/coordinates)	Current only	Yes	Open API (skolverket.se)
Auxiliary	Geocoded population register: student-dwelling linkage, RTB (Statistics Sweden, SCB)	Individual-level, geocoded dwelling address	TBD	No	Microdata service (MONA)

[1] Previously used by Lindgren (2021).

[2] School-level statistics were suppressed, not declassified, from Sept. 2020: a Dec. 2019 Kammarrätten i Göteborg ruling that independent-school throughput/grade data could be a trade secret led Skolverket to stop publishing any school-unit-level statistics, public or independent. A temporary law (July 2021, initially through June 2023) restored publication retroactively; school-unit-level data is confirmed live today via the PxWeb API, though the permanent legal basis post-2023 was not independently verified. Since spring 2025, small-cell suppression applies specifically to *staff* counts (<3 FTE at a school); no equivalent grade/test-result small-cell rule was found.

Data

Exposure

We identify exposure to noise pollution using the staggered rollout of mitigation technology: Trafikverket, the Swedish transportation authority, provides dwellings and neighborhoods with sound insulation and noise barriers, respectively. The BoViT program covers all provided insulation and barriers, including the exact dwellings treated and the date of provision. Data on insulation are, however, not publicly available and must be requested from Trafikverket. A smaller, publicly available subset covers only noise barriers, and construction dates are missing for many of these. Supplementary, publicly available data on dwelling, road, and rail positions let us determine which dwellings are located close enough to a road or railway to plausibly be eligible for treatment – i.e., to construct the sample of at-risk dwellings used in the staggered rollout design – and we can additionally request modeled background noise exposure and historical traffic density to use as controls.

Outcomes

We plan to draw on outcomes that can be measured with administrative diagnosis or register data rather than self-reports, spanning the life course. In adolescence, National Patient Register diagnosis codes would let us study mental health, sleep disorders, and insomnia. At birth, the Medical Birth Register would provide birthweight and other neonatal health measures for children exposed to noise in utero. In adulthood, the National Patient Register would let us study hypertension and, potentially, other diagnosed cardiovascular conditions, following Lindgren (2021).

Alongside health, students' learning outcomes will be shaped by noise pollution they are exposed to at school and at home. With publicly available school-level data, we can identify the effects of noise barriers next to school buildings. Individual-level test scores from Skolverket, matched via the geocoded population register (RTB), would let us identify the effects of home noise insulation using a more nuanced control group. Finally, access to the labor-market register would let us identify longer-term outcomes.

Potential Identification Strategies

- **Noise barriers (à la Moretti & Wheeler):** exploit the staggered construction of road/rail noise barriers, comparing dwellings and schools just inside vs. just outside the newly shielded area – separately for home exposure (residential DiD) and school exposure (school-level DiD using national test data), following Moretti and Wheeler (2025).
- **Sound insulation (à la Lindgren):** exploit the staggered timing of BoViT insulation as in Lindgren (2021), but compare treated dwellings to not-yet-treated eligible dwellings rather than to never-treated ones.

In either setting, we may use the timing of insulation or barrier construction relative to a child's birth or school-entry date to construct historical exposure measures to isolate in-utero and early-childhood exposure windows specifically. Across these designs, our main controls are the most important observable confounders: dwelling/school fixed effects, cohort or year-of-birth fixed effects, and family socioeconomic status from the population register.

References

Allroggen, Florian et al. (Oct. 2025). *Planes Overhead: How Airplane Noise Impacts Home Values*. w34431. Cambridge, MA: National Bureau of Economic Research, w34431. DOI: 10.3386/w34431.

Argys, Laura M., Susan L. Averett, and Muzhe Yang (Sept. 2020). "Residential Noise Exposure and Health: Evidence from Aviation Noise and Birth Outcomes". In: *Journal of Environmental Economics and Management* 103, p. 102343. ISSN: 00950696. DOI: 10.1016/j.jeem.2020.102343.

Basner, M. (June 2008). "Nocturnal Aircraft Noise Exposure Increases Objectively Assessed Daytime Sleepiness". In: *Somnologie - Schlafforschung und Schlafmedizin* 12.2, pp. 110–117. ISSN: 1432-9123, 1439-054X. DOI: 10.1007/s11818-008-0338-8.

Basner, Mathias and Sarah McGuire (Mar. 14, 2018). "WHO Environmental Noise Guidelines for the European Region: A Systematic Review on Environmental Noise and Effects on Sleep". In: *International Journal of Environmental Research and Public Health* 15.3, p. 519. ISSN: 1660-4601. DOI: 10.3390/ijerph15030519.

Beghelli, Silvia, Augustin De Coulon, and Mary O'Mahony (Oct. 2023). "Health Benefits of Reducing Aircraft Pollution: Evidence from Changes in Flight Paths". In: *Journal of Population Economics* 36.4, pp. 2581–2607. ISSN: 0933-1433, 1432-1475. DOI: 10.1007/s00148-023-00964-y.

Bodin, Theo et al. (Jan. 29, 2015). "Annoyance, Sleep and Concentration Problems Due to Combined Traffic Noise and the Benefit of Quiet Side". In: *International Journal of Environmental Research and Public Health* 12.2, pp. 1612–1628. ISSN: 1660-4601. DOI: 10.3390/ijerph120201612.

Boes, Stefan, Stephan Nüesch, and Steven Stillman (Sept. 2013). "AIRCRAFT NOISE, HEALTH, AND RESIDENTIAL SORTING: EVIDENCE FROM TWO QUASI-EXPERIMENTS". In: *Health Economics* 22.9, pp. 1037–1051. ISSN: 1057-9230, 1099-1050. DOI: 10.1002/hec.2948.

Clark, Charlotte et al. (2010). "The Effects of Road Traffic and Aircraft Noise Exposure on Children's Episodic Memory: The RANCH Project". In: *Noise and Health* 12.49, p. 244. ISSN: 1463-1741. DOI: 10.4103/1463-1741.70503.

Dean, Joshua T. (Oct. 1, 2024). "Noise, Cognitive Function, and Worker Productivity". In: *American Economic Journal: Applied Economics* 16.4, pp. 322–360. ISSN: 1945-7782, 1945-7790. DOI: 10.1257/app.20220532.

Greenhill, Simon (2026). *Noise Pollution and Infant Health*. URL: https://simondgreenhill.github.io/Greenhill_Noise.pdf.

Griefahn, Barbara, Anke Marks, and Sibylle Robens (Aug. 2006). "Noise Emitted from Road, Rail and Air Traffic and Their Effects on Sleep". In: *Journal of Sound and Vibration* 295.1–2, pp. 129–140. ISSN: 0022460X. DOI: 10.1016/j.jsv.2005.12.052.

Haines, M. M. et al. (Feb. 2001). "Chronic Aircraft Noise Exposure, Stress Responses, Mental Health and Cognitive Performance in School Children". In: *Psychological Medicine* 31.2, pp. 265–277. ISSN: 0033-2917, 1469-8978. DOI: 10.1017/S0033291701003282.

Hener, Timo (Nov. 2022). "Noise Pollution and Violent Crime". In: *Journal of Public Economics* 215, p. 104748. ISSN: 00472727. DOI: 10.1016/j.jpubeco.2022.104748.

Hygge, Staffan, Gary W. Evans, and Monika Bullinger (Sept. 2002). "A Prospective Study of Some Effects of Aircraft Noise on Cognitive Performance in Schoolchildren". In: *Psychological Science* 13.5, pp. 469–474. ISSN: 0956-7976, 1467-9280. DOI: 10.1111/1467-9280.00483.

Klatte, Maria et al. (May 2017). "Effects of Aircraft Noise on Reading and Quality of Life in Primary School Children in Germany: Results From the NORAH Study". In: *Environment and Behavior* 49.4, pp. 390–424. ISSN: 0013-9165, 1552-390X. DOI: 10.1177/0013916516642580.

Krekel, Christian, Johannes Rode, and Alexander Roth (2023). *Do Wind Turbines Have Adverse Health Impacts?* IZA Discussion Papers 16505. Bonn: Institute of Labor Economics (IZA). HDL: 10419/282632. URL: https://hdl.handle.net/10419/282632.

Krekel, Christian and Alexander Zerrahn (Mar. 2017). "Does the Presence of Wind Turbines Have Negative Externalities for People in Their Surroundings? Evidence from Well-Being Data". In: *Journal of Environmental Economics and Management* 82, pp. 221–238. ISSN: 00950696. DOI: 10.1016/j.jeem.2016.11.009.

Lercher, Peter, Gary W. Evans, and Markus Meis (Nov. 2003). "Ambient Noise and Cognitive Processes among Primary Schoolchildren". In: *Environment and Behavior* 35.6, pp. 725–735. ISSN: 0013-9165, 1552-390X. DOI: 10.1177/0013916503256260.

Li, Liming et al. (May 2021). "Aircraft Noise Control Policy and Mental Health: A Natural Experiment Based on the Longitudinal Aging Study Amsterdam (LASA)". In: *Journal of Epidemiology and Community Health* 75.5, pp. 458–463. ISSN: 0143-005X, 1470-2738. DOI: 10.1136/jech-2020-214264.

Lindgren, Samuel (Oct. 2021). *A Sound Environment: Health Effects of Traffic Noise Mitigation*. Working Papers 2021:10. Swedish National Road & Transport Research Institute (VTI). URL: https://ideas.repec.org/p/hhs/vtiwps/2021_010.html.

Moretti, Enrico and Harrison Wheeler (Sept. 2025). *The Traffic Noise Externality: Costs, Incidence and Policy Implications*. w34298. Cambridge, MA: National Bureau of Economic Research, w34298. DOI: 10.3386/w34298.

Nyberg, Erik (Nov. 2025). *Air and Noise Pollution: A State-of-the-Art Review of Causal Health Outcomes*. Working Papers 2025:6. Swedish National Road & Transport Research Institute (VTI). URL: https://ideas.repec.org/p/hhs/vtiwps/2025_006.html.

Trimmel, Michael et al. (Nov. 2012). "Effects of Low Intensity Noise from Aircraft or from Neighbourhood on Cognitive Learning and Electrophysiological Stress Responses". In: *International Journal of Hygiene and Environmental Health* 215.6, pp. 547–554. ISSN: 14384639. DOI: 10.1016/j.ijheh.2011.12.007.

Van Praag, Bernard M. S. and Barbara E. Baarsma (Jan. 1, 2005). "Using Happiness Surveys to Value Intangibles: The Case of Airport Noise". In: *The Economic Journal* 115.500, pp. 224–246. ISSN: 0013-0133, 1468-0297. DOI: 10.1111/j.1468-0297.2004.00967.x.

Wang, Scarlett Sijia et al. (May 2022). "Impact of Aeroplane Noise on Mental and Physical Health: A Quasi-Experimental Analysis". In: *BMJ Open* 12.5, e057209. ISSN: 2044-6055, 2044-6055. DOI: 10.1136/bmjopen-2021-057209.

Zou, Eric Yongchen (2017). *Wind Turbine Syndrome: The Impact of Wind Farms on Suicide*. URL: https://api.semanticscholar.org/CorpusID:160015504.